# Graph takeaways and explanations

[Website link]

## 01: Intro and setting the problem space

### 1a. Share of flights with and without delays in the US

This is a stacked bar graph of the number of flights that were on time (blue) and delayed (red). For every year except 2020 (due to COVID where travel was all time low), you can see that at least 20% of arrivals had some sort of air traffic incident involved whether it be delays, cancellations, or diversions. While 20% might not seem super high in a general perspective, considering the number of flights that take place on a yearly basis 20% is a considerable issues for the airline industry in the US. Not only does this lead to customer dissatisfaction but also leads to immense profit loss within companies.

### 1b. Yearly progression of arrival delays of 10 most delayed carriers

This is a stacked line plot the number of delays the top 10 most delayed carriers had each year from 2013 to 2023. For every year except 2020 (due to COVID where travel was all time low), you can see that at least 20% of arrivals had some sort of air traffic incident involved whether it be delays, cancellations, or diversions. While 20% might not seem super high in a general perspective, considering the number of flights that take place on a yearly basis 20% is a considerable issues for the airline industry in the US. Not only does this lead to customer dissatisfaction but also leads to immense profit loss within companies.

### 1c. Top 20 airports with most number of arrival delays

Below is a bar chart of the top 20 airports that had the most number of delayed flights. Unsurprisingly, major international hubs like Atlanta (ATL), Chicago O'Hare (ORD), and Dallas-Fort Worth (DFW) rank at the top, as these airports handle an enormous volume of flights daily. Interestingly, some mid-sized airports also appear in the top 20, likely due to regional congestion or weather-related disruptions. The chart reinforces the idea that the busiest airports tend to have the most delays, but other factors like geography and airline scheduling inefficiencies also play a significant role in delay patterns.

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## 02: Flight delays through the years and between states

### 2a. Minutes of arrival delays by state

This is a choropleth of each state's number of minutes of flight delays. As expected, states like California and Texas which are heavily populated and also have major well-known airports that serve millions of passengers annually have the highest amount of delays. To our surprise, Illinois is indicated to be the 4th highest overall delay concentrated state compared to a state like New York. Interestingly enough, these states California, Texas, Illinois, and Florida are states with popular national airports but also lots of extreme weather conditions which makes sense why these states are delay prone with air flights.

### 2b. Percentages of total arrival delays for top and bottom 10 states

Below are the top and bottom 10 states in arrival delays and their percentage values. This is just another contribution towards the choropleth map we made earlier further showing how states like Texas, Florida, and California make up 30%+ of delay data alone compared to the other remaining states. Looking at the bottom 10 graph, you can see how states that are known to have smaller population have low delay data and collectively don't even make up 20% of delays in the USA. A good amount of these states in the top 10 are also tourist attraction spots or areas with popular universities which also contribute to more flight delays.

### 2c. Arrival delays through the years, by state

This is a choropleth of each state's number of delayed flights received, through the years. California is constantly the number 1 in air traffic except 2020, which is due to covid when international regulations tightened for flights, and a huge amount of California flights are international. On the other hand, Texas has more domestic travels which is why 2020 there is more air traffic. For most other states other than the 4 mentioned in the overall graph, there isn't much change in air traffic throughout the years with exceptions such as North Carolina or Georgia having increases in travel for 1 year then going back to their average delay.

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## 03: Patterns in flight delays and other factors

### 3a. Reasons for flight delays, by region

This is a stacked bar chart of different reasons for or types of flight delays, separated by geographical region in the US. We can see that similar to the low correlation in the heatmap, security delay doesn't even make up 10% of recorded delays for any region. Interestingly though, the weather doesn't seem too big of a contributor just based off this stacked bar chart. The Midwest and West Coast have the biggest stacks for weather which makes sense as these areas of America have states with more extreme weather conditions. Similar to the heatmap results, carrier, NAS, and late aircraft delays make up approximately 80% of all delay data for each region. This further proves that operational efficiencies like scheduling, maintenance, and carrier management are the biggest issues that contribute delays nationwide and this is a big issue especially in the West coast as seen through the tallest bar within this chart.

### 3b. Correlation between different causes of delays

This is a heatmap of the correlation between the different causes of delays. We see when comparing air delay (minutes) to other specific delay features, the correlation between late aircraft is 0.96, carrier is 0.94, NAS 0.86, weather 0.77, and security 0.38. While it seems security doesn't have much impact on airline delays, it seems that features like weather and NAS have a moderately high contribution towards higher delay times. Late aircrafts have 0.96 correlation to arrival delays which indicates that late delays from previous flights have a huge impact on arrival delays. These results suggest that while airline control factors like carrier and aircrafts have the most significant affect towards delay increases, systemic issues like traffic congestion weather conditions also have almost as significant impact. This indicates the fact that several adjustments are required for improvement whether it be operational improvements, air traffic management, enhanced weather forecasting strategies, etc.

### 3c. Relationship between number of delayed flights and total US income, by year

This is a scatter plot of the relationship between the number of delayed flights and the total US income. You could also see there is an outlier of 2020 which is because of COVID where air travel regulations were strict and economic activity was at an all-time low. There is notable clustering of data within the upper right side of the data which indicates that though not for every increase in delay count, higher delay counts generally lead to moderate high income. This was interesting because initially we had 100% expected income levels to drop as more delays typically would be expected to drop income levels. However, thinking in a broader economic perspective, more delays leads to the fact that there is lots of air traffic which means strong economic activity. Even though there might be so many delays that could lead to income loss, these datapoints suggest that high economic activity sort of outweighs those losses from delays.

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## Summary of findings

After analyzing flight delays and their causes, we uncovered some interesting patterns. One major takeaway is that American Airlines and Southwest Airlines had the most arrival delays, while JetBlue, Delta, and United had significantly fewer. This shows that choosing certain airlines over others might make a difference if people want to avoid delays. We also found that Florida, California, and Texas had the highest number of arrival delays and delays per minute. This makes sense given that they are the 3 most populated states. However, this dataset alone isn't likely to change passenger behavior, as people will continue flying to these major hubs regardless of potential delays. On the other end of the spectrum, states like Alabama, New Mexico, and Nebraska had the lowest share of arrival delays, reinforcing the idea that larger, busier states tend to experience more disruptions.

One surprising finding was that New York didn't rank among the top states for delays, even though it's the fourth most populated state and home to New York City. Our D3 visualization showed NYC as one of the most delayed airports, yet our heatmap of state-level delays didn't reflect the same trend. This suggests that delays in New York may be concentrated at specific airports rather than being a statewide issue. The same could be applied to Illinois as well as we found that Chicago had the most delays but Illinois didn't come out as the most delayed city.

To help reduce delays, airports in high-delay areas might consider upgrading air traffic management systems or expanding runway capacity to handle growing demand. On the passenger side, travelers can choose less busy airports or book flights during off-peak hours to reduce the risk of delays. Looking ahead, future research could dig deeper into seasonal trends, weather impacts, and peak travel periods to provide more specific insights. Using machine learning models to predict delays based on past data could help airlines and airports prepare for disruptions before they happen. Expanding the dataset to cover international flights and a longer period could also give a more complete picture of flight delay trends.